LAM NGO

Espoo, Finland • Email • Scholar • LinkedIn • GitHub • Website

Research Interests: Digital Twins, 3D Computer Vision, AI for Science & Health

EDUCATION

PhD in Electrical Engineering **09/2026 - Present**
ELLIS Institute Finland & Aalto University *Espoo, Finland*

- Research focus: anatomical digital twins from medical imaging
- Advisor: Jiancheng Yang

Master of Science in Computer Science **09/2023 - 12/2024**
City St George's, University of London • *Grade: Distinction* *London, UK*

- Thesis: Feature Extraction from Deep Dynamic Network for Reinforcement Learning
- **Global STEM Leadership Master's Scholarship**: 75% tuition award, given to 10 top applicants

Bachelor of Science in Mathematics, Minor in Biology **08/2016 - 05/2020**
Sewanee: The University of the South • *GPA: 3.46 / 4.0* *TN, USA*

- **International Honors Scholarship:** partial tuition award

RESEARCH EXPERIENCE

Doctoral Researcher **09/2026 - Present**
ELLIS Institute Finland & Aalto University *Espoo, Finland*

Project: Simulation-Ready Mesh (in progress)

- Leading research on reconstructing simulation-ready anatomical meshes from sparse observations using score-based diffusion models.

Project: Pulse-agent (in progress)

- Leading research on agentic workflows for automated CT image segmentation.

Project: DART – ICLR 2027 (under review)

- Collaborated with Johns Hopkins University and contributed baseline experiments to DART, a shared shape representation that corrects segmentation errors using anatomical text descriptions without access to images, with up to 31% Dice gains on individual cases across 13 public benchmarks.

Project: FINe-Health Foundry – Business Finland

- Contributing to a Business Finland-funded project developing a national AI foundation model for healthcare decision-making.

AI Research Engineer **09/2024 - 06/2026**
City St George's, University of London *London, UK*

Project: Federated Dataset Simulation – NeurIPS 2026

- Developed a VLM- and LLM-guided pipeline that splits image datasets into federated clients by visual traits instead of class labels; the splits are up to 2.9× more diverse than Dirichlet label skew and extend to segmentation and detection.
- Benchmarked federated training on 4 datasets (classification, segmentation, detection; 5–20 clients): the splits create a tougher stress test than the standard label skew, lowering accuracy by up to 9.9 points vs. 8.0.

Project: DeeP-Mod – ISNN 2025

- Researched and developed methods for training reinforcement learning agents effectively in noisy game environments.

INDUSTRY EXPERIENCE

C++ Programmer Mentee **10/2023 - 03/2024**
Ubisoft Leamington *Leamington Spa, UK*

- Built a graphics engine using modern C++ and internal graphics API.
- Implemented AI algorithms to anticipate player movements, enhancing gameplay difficulty.

Junior Programmer **06/2022 - 06/2023**
Indi Games Inc. *Ho Chi Minh City, Vietnam*

- Developed game systems in Unity and C# with real-time WebSocket integration for data-driven gameplay.

PUBLICATIONS

- Siomos, V., **Ngo, L.**, Passerat-Palmbach, J. & Tarroni, G. Federated Dataset Simulation: Inducing Label-Free Heterogeneity Across Tasks. *NeurIPS 2026 Evaluations and Datasets Track.*
- Swarna, K., Bassi, P., **Ngo, L.**, Li, W., Yang, J., Yuille, A. & Zhou, Z. DART: Scaling a Shape Representation Across Anatomical Classes and Imaging Modalities. *Under review at ICLR 2027. Preprint: arXiv:2610.27605*
- Child, C. & **Ngo, L.** DeeP-Mod: Deep Dynamic Programming Based Environment Modelling Using Feature Extraction. *ISNN 2025, Springer LNCS vol. 15951. doi:10.1007/978-981-95-1233-1_3*

PRESENTATIONS

- Presented *DeeP-Mod* at the 19th International Symposium on Neural Networks (ISNN 2025)

SKILLS

Programming languages: Python, C++, C#

ML: PyTorch, PyTorch3D, TensorFlow, Keras, CUDA, NumPy, Pandas, Scikit-learn, NVIDIA FLARE

Technology: SLURM, Git, Linux, Multi-GPU/Multi-Node Training, FastAPI, Docker

LLMs & VLMs: Claude, GPT, InternVL3.5, Llama, Qwen3, CLIP

Agentic AI: Claude Code, OpenAI Agents SDK, Claude Agent SDK, MCP